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100%

Module #	671101000-KM-07:
NQF Level	level 4,
Notional hours	310
Credit(s)	Cr 31
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

SUMMATIVE ASSESMENT

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Ziphozihle Mqopi	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Ziphozihle Mqopi

Learner's signature: _____ Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

_____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Ziphozihle Maqopi	Signature
Assessors Name:	Signature:

EASTC TECHNOCENTRIC VARSITY

LEARNING UNIT 1:

KM-07-KT01: Concepts, theories and principles of supply Systems (20%)

Learning Outcome 1:	KT0101 Theories and concepts of alternating current
Learning Outcome 2:	KT0102 Fundamental principles of alternating current
Learning Outcome 3:	KT0103 Alternating current generation (distribution systems theory)
Learning Outcome 4:	KT0104 Characteristics and calculations from alternating current waveforms
Learning Outcome 5:	KT0105 Theories, concepts and principles of direct current sources
Learning Outcome 6:	KT0106 Sources of direct current
Learning Outcome 7:	KT0107 Calculations of direct and alternating current circuits
Learning Outcome 8:	KT0108 Construction and operating principles of direct current generators
Learning Outcome 9:	KT0109 Characteristics, sources and generation of renewable energy
Learning Outcome 10:	KT0110 Relevant legislative requirements

Task	Questions Description	Marks
1	a) Explain the principles of generation by using Fleming's right hand rule and Faraday's laws	4
	b) Explain the load balancing principles in a three phase supply system.	4
	c) Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account	4
	d) Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods.	4
	e) Describe with the aid of drawings the components of direct current generators and their functions.	4

LEARNING UNIT 2: Batteries KM-07-KT02: (10%)

Learning Outcome 1:	KT0201 Fundementals of battery operation
Learning Outcome 2:	KT0202 Types and classification of batteries
Learning Outcome 3:	KT0203 Safety precautions of batteries
Learning Outcome 4:	KT0204 Care and maintenance of batteries
Learning Outcome 5:	KT0205 Battery components and construction
Learning Outcome 6:	KT0206 Disposal of batteries

Task	Questions Description	Marks
2	a) Explain the terms potential difference, electromotive force, relative density and capacity	3
	b) Describe and explain the various types of batteries and cells	3
	c) Describe and explain the correct procedures to care, maintain, store and dispose of batteries and the hazards and safety precautions associated with batteries and battery rooms	4

2a) Pontential Difference: Voltage between two points in a circuit. Energy used per coloumb by a component unit volts (V)
Electromotive force: Total energy supplied per ~~col~~ coulomb by a source like a cell. ~~Source~~ or generator. It is the voltage produced by the source when no current flow. It is measured in volts (V)
Relative Density: It is the ratio of a substance's density to water's density. Used to test battery electrolyte
Capacity: Total charge a battery can supply. It is measured in ampere-hours (Ah)
b) Primary cells: are non-rechargeable and designed for one-time use because their chemical reaction cannot be reversed.

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LEARNING UNIT 3:

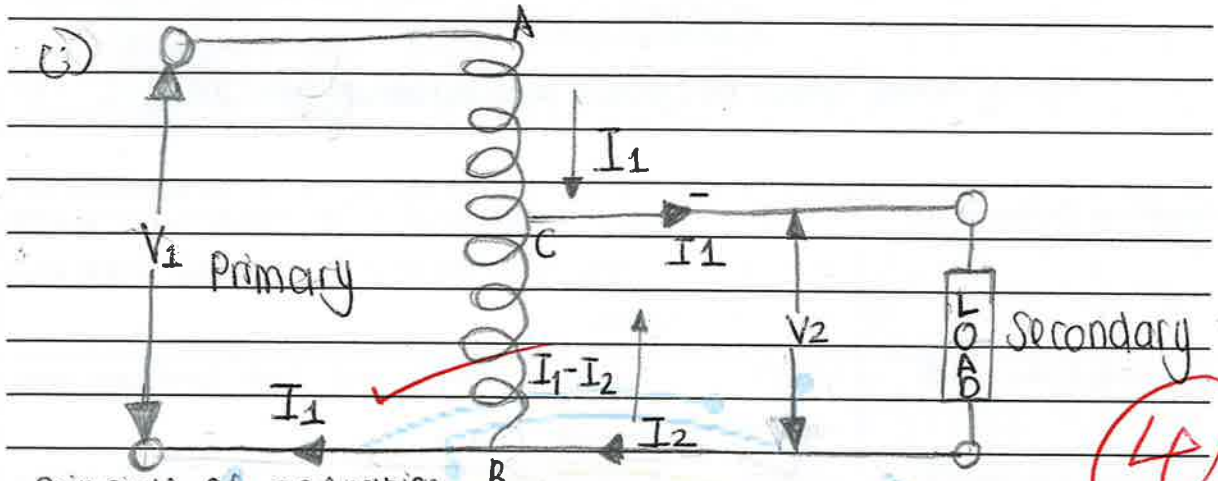
KM-07-KT03: Transformers (20%)

Learning Outcome 1:	KT0301 Theories of single phase wound transformers
Learning Outcome 2:	KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.
Learning Outcome 3:	KT0303 Fundamentals of transformer construction
Learning Outcome 4:	KT0704 Transformer cooling systems
Learning Outcome 5:	KT0305 Principles of single phase transformer operation
Learning Outcome 6:	KT0306 Principles of single phase auto-transformer operation
Learning Outcome 7:	KT0307 Transformer losses
Learning Outcome 8:	KT0308 Formulas and calculations on input and output of transformers
Learning Outcome 9:	KT0709 Maintenance requirements of transformers
Learning Outcome 10:	KT0310 Types of three phase transformers and their applications
Learning Outcome 11:	KT0311 Construction of three phase transformers including open core, closed core and shell or divided core
Learning Outcome 12:	KT0312 Principles of 3 phase transformer operation and configuration
Learning Outcome 13:	KT0313 Principles of 3 phase auto transformer operation and configuration
Learning Outcome 14:	KT0314 Transformer protection and phasing

Task	Questions Description	Marks
3	a) Name and describe the types of single phase transformers	3
	b) Describe and explain the cooling systems used on transformers	4
	c) Explain, with the aid of drawings, the principle of operation of single phase auto- transformers	4

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(i)



Principle of operation

1. Single Shared Winding: Unlike standard transformers that have 2 physically 'seperated coils, an autotransformer uses one coil Part of this coil serves as the primary (input) winding while a portion serves as a secondary (output) winding
2. Electromagnetic Induction: When an AC supply is applied to the primary section, it creates alternating magenetic flux in the iron core. This change in magnetic flux induces a voltage across the entire winding.
3. Conductive and Inductive Transfer: Power is transferred to the secondary load in two ways. A portion of the energy is transferred inductively through the magnetic field between the coils and the rest is transferred conductively directly through the physical electrical connection on the shared coil
4. Voltage Transformation Ratio: The output voltage depends on the numbers of turns tapped. The relationship is proportional to the turns ratio $\frac{V_1}{V_2} = \frac{N_1}{N_2}$ where V is voltage and N is the number of turns.

(4)

d) Regular inspection should be carried out to check for oil leaks, damaged brushings and base connections because these faults can cause overheating on the transformer. Oil testing and filtering must be done periodically on the transformer to monitor moisture, acidity and dielectric strength since contaminated oil reduces insulation and cooling efficiency. Cleaning and tightening of terminals of the transformer, cooling fins and breathers is required to prevent dirt buildup and corrosion on the transformer

(3)

(2)

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2. Coaxial Cable:

Characteristics: Has a central copper conductor surrouded by insulation, a metal shield and an outer cover

Applications: used for cable television, broadband internet.

Advantage: offers better shielding than twisted pair and performs well over moderate distances

Disadvantage: Bulkier than other cables and becoming less common in modern networks

3. Fiber Optic Cable:

Characteristics: Transmits data using light through thin glass or plastic fibers. Comes in single-mode and multi-mode types

Applications: Ideal for ISP backbones, data centers and long-distance communications

Advantages: Delivers very high speeds, covers long distances and is immune to interference

Disadvantages: More expensive and requires specialized equipment and expertise to install

4 USB Cable: Characteristics: Contains multiple wires for both data transfer and power delivery. Application: connects peripheals like printers, drives, and phones and is used for charging. Advantges. Universally compatible supports plug-and-play, and carries both data and power.

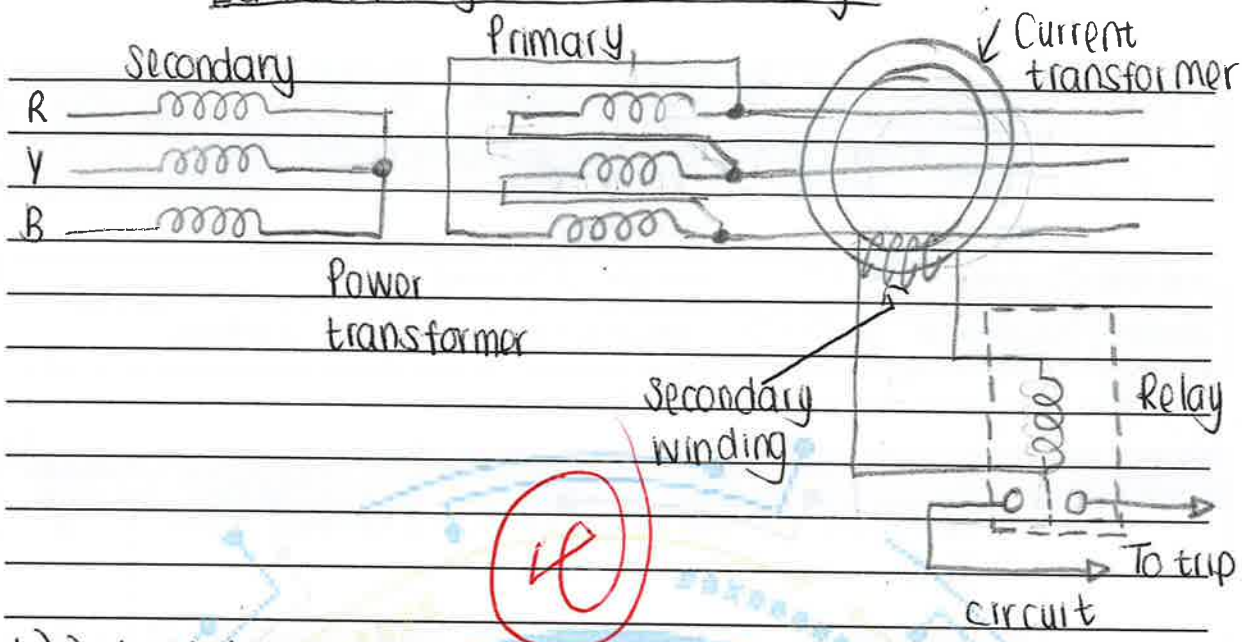
Disadvantages. Limited to short distance typically under 5 meters.

b) SANS 10142-1 states that cables must be correctly sized for voltage and current to prevent overheating and fire. They must be suited to the enviroment, with proper insulation and protection against heat, moisture, and mechanical damage using conduit or armouring where needed. Correct installation methods must be followed, including proper support, bending radii and secure terminations. Joints are not permitted in inaccessible places and power cables must be separated from data cables to avoid interference. All cabling must be properly earthed and installed to ensure the safety of people, property and equipment

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Earth Leakage Protection Relay



b) i) Moulded Case: The circuit breaker is housed in a factory-moulded insulating case made of high-strength plastic. This protects internal components, provides electrical insulation, and makes the unit compact and robust for panel mouting

2) Positive Indication: The breaker clearly shows its true contact status through handle position or an indicator window. "On", "OFF" and "Tripped" positions are mechanically linked to the contacts so what you see reflects the actual state

3) Trip Positions: A distinct mid-position of the handle that indicates the breaker has tripped due to a fault/overload. It is seperate from "On" and "OFF" and the breaker must be reset by moving the handle fully to "OFF" before switching back "ON"

4) Factory Sealed: The calibration and trip settings are set and sealed by the manufactures. The unit cannot be opened or adjusted on site, ensuring reliable and consistent protection

5) Thermal Magnetic Tripping: Uses two elements. The thermal element responds to sustained overloads using a bitmetallic strips that bend with heat

6) Quick Break: The contacts open rapidly regardless of how slowly the handle is moved. This snap-action minimizes arcing time, reducing contact wear and improving safety.

7) Trip Free Mechanisms: The breaker will trip even if the handle is held in the "On" position. This prevents overriding a fault and ensures the breaker opens under fault conditions.

8.) Interlope Barriers: Insulating barries fitted between poles / phase inside the breaker. They increase creepage and clearance distances to prevent flashover between phases during faults.

c) A C motors are protected by using overload relays and Circuit breakers. Thermal or electronic overload relays detect sustained overcurrent from overloads and disconnect the motor to ~~prent~~ prevent winding damage. Circuit breakers or fuses provide short-circuit and fault protection by instantly interrupting high fault currents.

Motor protection may also include under-voltage and phase-failure relay to prevent damage when supply conditions are abnormal.

d)

Time Delay: Overload devices use a time delay to allow brief, expected overcurrents like motor start-up without tripping. This prevents nuisance trips from temporary inrush currents while still tripping if the overload is sustained, protecting windings from overheating.

Current Rating: The device's current rating must match the motor's full load current. A rating too low causes unnecessary trips during normal operation. A rating too high fails to trip during actual overloads or faults, risking motor damage. Correct selection ensures reliable protection for locked rotors, and coordinates with fuses or breakers for short circuits.

e)

Advantages — Disadvantages

Single Phase Motor: — Three Phase Motor:

e)	Advantages	Disadvantages
	Single Phase Motor: Easy to install with basic wiring	Single Phase Motor: Limited to small installation only
	Three Phase Motor: Suitable for large complex systems	Three Phase Motor: Not easy to install needs skilled electrician

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	Ziphozinle Maqopi
Signature of Assessor	Signature of Candidate

Candidate Feedback Report

Candidate's Name	Ziphozinle Maqopi	ID No.	0802030118086
Assessor's Name		Reg. No.	

ASSESSMENT DECISION

	C	NY C	Comments

OVERALL COMPETENCE JUDGEMENT:

Comments	
Date	
Signature of Assessor	Signature of Candidate

Candidate's Name	Ziphozinle Maqopi	ID No.	0802030118086
Assessor's Name		Reg. No.	

Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

ASSESSMENT REVIEW

NAME of LEARNER	Ziphozinle Mqopi	NAME of ASSESSOR	
VENUE		DATE of REVIEW	
EASTC TECHNOCENTRIC VARSITY			

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐Agree ☐Disagree	☑Agree ☐Disagree	
The assessment related to the registered unit standard?	☐Agree ☐Disagree	☑Agree ☐Disagree	
The assessment was practical?	☐Agree ☐Disagree	☑Agree ☐Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐Agree ☐Disagree	☑Agree ☐Disagree	
The assessment instruments were fair, clear and understandable	☐Agree ☐Disagree	☑Agree ☐Disagree	
The assessment judgements was made against set requirements	☐Agree ☐Disagree	☑Agree ☐Disagree	
The venue and equipment was functional?	☐Agree ☐Disagree	☑Agree ☐Disagree	
Special needs were identified and the assessment plan was adjusted	☐Agree ☐Disagree	☑Agree ☐Disagree	
Feedback was constructive against the evidence required	☐Agree ☐Disagree	☑Agree ☐Disagree	
An opportunity to appeal was given	☐Agree ☐Disagree	☑Agree ☐Disagree	
The evidence was recorded	☐Agree ☐Disagree	☑Agree ☐Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING

I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid

Learner	Date	Assessor	Date	Moderator	Date

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